Drone-Based Intelligent Spraying of Pesticides: Current Challenges and Its Future Prospects

Chapter · March 2024

DOI: 10.1007/978-981-99-8684-2_12

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ISBN 978-981-99-8683-5 ISBN 978-981-99-8684-2 (eBook)
https://doi.org/10.1007/978-981-99-8684-2

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The registered company address is: 152 Beach Road, #21-01/04 Gateway East, Singapore 189721, Singapore

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Contents

12 Drone-Based Intelligent Spraying of Pesticides: Current Challenges and Its Future Prospects

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Abstract

With the advancement in drone technology over the past few years, drones have become an essential component of modern agriculture. Beginning with the first idea of an unmanned bomb-filled balloon to today's unmanned multi-rotor aerial vehicle, drone technology has witnessed big and rapid developments. The ongoing research on drones has led to a path where one after another modification in every aspect, be it an array of sensors or algorithms and the marketability of dif-

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S. S. Chouhan et al. (eds.), *Applications of Computer Vision and Drone Technology in Agriculture 4.0*, https://doi.org/10.1007/978-981-99-8684-2_12

ferent types and models of drones suitable for various agricultural operations from collecting data of crop parameters and field mapping to detecting pests and applying pesticides as and when required, has increased. Modern-day drones have advanced sensors and high-resolution cameras which, along with good processing algorithms, can distinguish between healthy, nutritionally deficient, disease-infected, insect-infested crop or an unwanted weed. The data collected can then be successfully utilized for crop modeling and searching and locating potential pest hotspots. Crop production is severely limited due to dramatic losses incurred by the farmers as a result of pests and diseases. Thus, the application of crop protection materials becomes one of the important practices in agriculture. However, the traditional methods of pesticide application are time consuming, less effective, as well as hazardous to human health. In this regard, sprayer-mounted drones have made pesticide application quick, efficient, and safe. Although cost, weather dependence, and other shortcomings limit drone use by farmers in general, the overall and long-term benefits provided by drones are that which establish drones as an integral part of modern farming systems. This holds scope for the future wherein drones might be seen as the face of hi-tech agriculture.

12.1 Introduction

With an increase in population and dwindling of resources, the focus of crop production has now shifted towards precision agriculture. The practical application of precision and hi-tech crop production is not always possible at a full extent in small-scale agricultural lands. However, large-scale or commercial growers are able to incorporate advanced technology in the agricultural field, as seen in developed countries where artificial intelligence-based machines have carried agricultural system to a higher level. The core of artificial intelligence (AI) is machine learning and deep learning, which makes problem-solving its main objective through processing algorithms. Hi-tech computer-based systems like drones have tremendously contributed to the agricultural sector. Initially, drones were used as military tools, but their broad application provided them with a powerful market opportunity. Drones have found their utility in precision agriculture, remote sensing, and photogrammetry. Semiautomated drones are especially helpful in agriculture planning, spatial information collection, and real-time data analysis. The drone technology underwent a long way of improvement and is still under modification to broaden its utility. Reduction in size, use of powerful sensors, processors, global positioning system (GPS) modules, and improved payload capability have made drones more applicable in agricultural systems. The current market is mostly based on semiautomated or small unmanned aerial vehicles (UAVs). UAVs are highly used for soil and field analysis, mapping, crop monitoring, crop height estimation, weeding, and spraying. A common application of drones in field is drone mounted with sprayer for pesticide application. Traditional method of pesticide spraying makes the laborers vulnerable to pesticide poisoning. Manual or mechanical spraying also fails to give full coverage, causes environmental pollution, is more labor dependent and time consuming, and has reduced effectiveness. Now, UAVs have made it possible to achieve a uniform application of pesticides over large areas, irrespective of rough terrain or labor availability. It has also reduced pesticide loss and application cost, saved time, and been more effective in controlling weeds, insects, and diseases by spraying crops according to requirement (Talaviya et al. 2020; Lost Filho et al. 2020; Borikar et al. 2022).

12.2 Development of Drone Technology for Spraying of Pesticides

Drones are officially defined as unmanned aircraft systems that are remotely pilotless. The wonderful part about drones is that they fly without a human or people inside controlling it as they soar through the skies. The human pilot who operates the drone on land using an onboard computer is occasionally referred to as an operator (Newcome 2004). Since it was first identified as a flying robot, drones are the most popular names for them. The term "drones" comes from honeybee colonies where the bees do their duties obediently (mindlessly) under the direction of a powerful queen located far away (Joiner 2018). Surprisingly, the drone functions similarly to bees in that its microcontrollers are configured by their operators to be on autopilot, and their technological innovation keeps growing.

As the timeline in the above-given data (Table 12.1) showed, in 1849, the Austrian balloons, often regarded as the first drone use, attacked Venice, Italy, with the help of a pilotless balloon loaded with bombs. Later, in World War I in 1916, the Ruston Proctor Aerial Target was utilized, followed by the Hewitt-Sperry Automatic Airplane, also known as the "flying bomb" (Valavanis et al. 2007), and the usage of drones advanced at that time. The level of innovation rose with the production of the Gipsy Moth in 1929 in the UK. With the installation of radio controls, a successor to the Gipsy Moth known as the de Havilland Tiger Moth was subsequently built. The DH82B Queen Bee, which was the first reusable and returnable UAV employed as practice targets, was the evolution's last iteration. Due to the widespread usage of drones in warfare, the Vietnam War was also impacted by the development of drone technology. Drones serve a variety of purposes, including carrying explosives,

Table 12.1 Timeline of the development of drones in the early years

Year	Drone name
1894	Austrian balloons
1916	Hewitt-Sperry Automatic Airplane "flying bomb"
1918	Kettering Bug
1929	Gypsy Moth
1935	DH.82B Queen Bee
1939	Radioplane
1950	TDD1-Target Drone Denny

acting as decoys in combat, shooting missiles at predetermined locations, and dispersing flyers for psychological warfare (Dalamagkidis et al. 2012). Target Drone Denny 1, also called TDD1, was a drone for which the US Navy secured a contract. They played the parts of radio-controlled model aeroplanes and actors. The term "drone" was first used in connection with an unmanned remotely piloted vehicle in TDD1 (Joiner 2018).

12.2.1 Advent of Drone Technology in Agriculture Sector

Since they were first created, aeroplanes have been extensively used for military operations. However, there was another disadvantage to aircraft: they were operated by people. However, the likelihood of an accident was great owing to any mechanical error, and high wind pressure caused planes to catch fire. Due to the possibility of such events, both property loss and loss of life were possible outcomes.

Abraham E. Karem is the creative brain behind the predator or drone. Abraham was born on June 27, 1937, in Baghdad, Iraq. After a few years, his family moved to Israel, where he was raised and attended college. He had a strong love for science and technology since he was young. He made the models for the aeroplane while he was only 14 years old. He conducted his tests in the garage of his residence. In addition, he earned a degree in aeronautics as a result of his love of science and flying machines. During the Second World War, he built his first drone for the Israeli Air Force. In 1970, he moved to California in the United States. Ibrahim may have resided in different nations, but he has always used his expertise and inventions to benefit the air force of each nation. The most well-known inventions of Abraham lacked more modern technology. It was employed by the Italian and American air forces. His first and most advanced drone, Amber, was also the most well-known predator drone. He is referred to as the Father of Drones as a result of his creation, Amber. He is also known as the inventor of unmanned aerial vehicles. Firstly, robotic plane entered the scene. These planes were piloted by robots. However, this kind of aircraft required a large expenditure because engineers had to handle the costs associated with both the robotic pilot and the drone. So, the price of that was high.

The first time the notion of creating something that could fly by itself and without humans came about was to overcome such a circumstance and save human life. Then came the development of unmanned aerial vehicles (UAVs) as they are more commonly known. UAVs operate remotely, and this technology has risen to the top for the air force. In the early 2000s, the first drone also known as unmanned aircraft was created for farming. These drones were primarily employed for mapping, crop monitoring, and crop spraying. Farmers could rapidly and precisely examine their fields, spot possible issues, and make data-driven decisions about planting, fertilizing, and harvesting thanks to their cameras and sensors. Early drones, however, were quite expensive and operated with a high level of technical skill. Drone use in agriculture has increased recently and is anticipated to do so in the future as technology improves and costs come down.

Since their inception, drones have been utilized in agriculture, although their use and application have evolved throughout time. Since the early twentieth century, aerial photography has been used in agriculture to survey and map farms, monitor crop development, and spot possible problems like pest infestations or irrigation concerns. Farmers are able to monitor enormous expanses of land more effectively and accurately than they could by strolling the fields because to the use of aircraft and satellites to take pictures of crops and fields.

12.3 Types of Drones Used for Spraying in Agriculture

Nowadays, semiautomated type of drones are being increasingly used in the agricultural sector. Unmanned aerial vehicle (UAV) is a small aircraft which is equipped with different kinds of camera, sensors, and controlling devices based on the purpose or application. There are different types of UAV models which are being used from the past decade as depicted in Fig. 12.1.

12.3.1 Fixed Wing

These rotor drones have stationary wings, which are fixed at their positions. The wings are like that of a plane or are aerofoil shaped. For takeoff, they need to be given inertia from an external source. When the drone reaches a certain speed, the wings create the lift. This type of drones are highly deployable and can stay up to 1 h in the air. They can cover a huge area in rapid speed. They are mostly suitable for geographical mapping and monitoring channel. Fixed-wing hybrid vertical take-off and landing (VTOL) is a new line of fixed-wing drones which integrate the features of a UAV to hover at one place. They can take off from one place and can sustain over an area for some time and have more flight endurance.

12.3.2 Single-Rotor Helicopter

This drone has a set of horizontally rotating blades at the top, which are attached to a central mast and another smaller sized rotor on the tail. The blades produce the thrust and lift the craft, allowing vertical takeoff and landing. Helicopters can fly forward and backward and hover at a particular place for long. They are highly deployable, have more endurance due to gas power system, and can be flown in

Fig. 12.1 Different types of UAVs: (**a**) fixed wing, (**b**) fixed-wing VTOL, (**c**) single-rotor helicopter, (**d**) quadcopter

winds blowing at 20–50 mph. Due to their long rotor blades, they have less spin and more efficiency and are hence advantageous in agricultural fields. They are more suitable than fixed-wing drones as they can be used in congested or remote areas. They are also used for scanning in aerial view and wide-range data transfer. However, their flight duration in the air is less than fixed-wing type.

12.3.3 Multi-Rotor Copters

Commercial spray drones are mostly multi-rotor types. These are rotorcrafts with multiple sets of horizontally rotating blades. The blades provide the lift and control over movements. Multi-rotors have a better command over positioning. They are also easy to use and control. They are generally used for taking top-view photography. However, they lack in speed and endurance, have little payload capacity, have less flight time, and have limited use in agricultural operations. Based on the number of rotors, multicopters can be distinguished as quadcopter (four blades), hexacopter (six blades), and octocopter (eight blades) (Talaviya et al. 2020; Mogili and Deepak 2018; Hafeez et al. 2022; Dileep et al. 2020).

When used for spraying of pesticides, drones use different technologies and algorithms like wireless sensor networks, gyroscope sensors, accelerometer sensors, Arduino, and spray motors (Garre and Harish 2018; Pharne et al. 2018; Yallappa et al. 2017). They utilize established methodologies for identification and spraying like telerobotic navigation and target selection and filter-paper ratio assured titration method for spraying droplets (Adamides et al. 2012; Shang et al. 2004; Zhenyu et al. 1996). For pesticide application, drones are equipped with suitable sprayers for their uniform application. The sprayer is used to crumble the liquid into tiny droplets and release it with very little power. For drones, four types of energy-based sprayers are used, namely hydraulic energy sprayer, gaseous energy sprayer, kinetic energy sprayer, and centrifugal energy sprayer (Talaviya et al. 2020). Just like conventional aerial sprayers, these are also equipped with the basic components like tank, pressure pump, hoses, filters, nozzles, and flowmeter. Besides these, they also have GNSS receiver and multiple sensors to avoid collision. Spraying system in a drone can have different types of nozzles depending on the technical specifications. UAVs use two techniques for spraying—either the spraying system is on a boom sprayer or the spraying system is under the propeller rotor. Nowadays, drones use rotary atomizers or controlled droplet atomizers to spray using very low pressure and produce uniform droplets (Hanif et al. 2022; Anonymous 2023).

12.4 Reasons for Using Drone Technology in Agriculture

Thousands of drones are currently being used throughout the world for a wide range of purposes, including military, photography, agricultural, and many others. Now drone use has moved beyond the military and recreational users to include businesses like agriculture. There are currently drones available on the market that were created especially for agricultural use. Farmers can benefit from using drones in agriculture because they have some clear advantages over traditional methods, including increased field capacity and efficiency, shorter turnaround times, fewer field operational delays, reduced pesticide and fertilizer waste due to high levels of atomization, water conservation due to ultralow-volume spraying technology, and lower spraying and fertilizer application costs. There are many features equipped in drones, which aid farmers in overcoming obstacles to improve crop yield. The agriculture sector can be made easily manageable for both large and small fields with the use of drones (Erin 2017).

Drones are expected to mark a significant shift in how people cultivate crops over time. By relying on drone technology, precision agriculture can be carried out while tackling food and water crises (Mogili and Deepak 2018). The management of a crop using big data and GPS is known as precision agriculture (Huuskonen and Oksanen 2018). Using their capabilities, drones have recently been assisting farmers in increasing crop productivity. For instance, drones may fly at a height of 120 m and capture, modify, and analyze each individual leaf on a maize plant. They can also collect data on soil water-holding capacity and variable-rate water application, providing agriculture intelligence to farmers and agricultural consultants (Khamuruddeen et al. 2019). The use of drones in agriculture is being developed to support both local and large farming enterprises. According to the Association of Uncrewed Vehicle Systems International, the legalization of commercial drones is expected to have a positive economic impact of more than €70 billion between the years of 2015 and 2025 (Veroustraete 2015). According to the forecast, precision agriculture would contribute significantly to economic expansion.

As can be seen, drone applications in agriculture include crop monitoring, aerial pesticide or fertilizer application, and a few more. The topic of aerial spraying is covered in 60% of the papers cited. Particularly in Asian nations like China, Japan, and South Korea, aerial spraying has many potential applications. Japan created UAV spraying in 1997 to treat thousands of hectares of crops and forestry. The pilotless helicopter, Yamaha R-MAX, was developed by Yamaha Corporation (Japan) for agricultural use. In order to prevent the helicopters from being utilized by others, exports were prohibited later in 2007 (Xue et al. 2016). Yamaha no longer sells Yamaha R-MAX, but instead provides services to interested farmers. On the other hand, China has transformed the concept of an unmanned helicopter into the DJI Agras MG-1, a foldable octocopter with a compact design, smart flight mode, and carbon fiber components. One of the big Chinese drone manufacturers, DJI has monopolized the industry by producing different sizes of drones, from tiny to powerful ones. Many academics created their unmanned drones for agricultural use as a result of inspiration from both nations. In Table 12.2 are listed the characteristics and details of the Yamaha helicopter, DJI drones, and other agricultural drones used by researchers.

Table 12.2 Features and specifications of drones for spraying in the agriculture sector

Properties	N-3 UAV	Agras MG-1 DJI	3WQF120–12	Yamaha RMAX	3CD-15	Drone-mounted sprayer	WSZ-0610	6-Rotor UAV	HY-B-15 L
Maximum speed	4 m/s	8 m/s	5 m/s	–	6 m/s	–	4 m/s	16 m/s	4.5 m/s
Flight time duration	–	–	50 min	1 h	30 min	–	30 min	25 min	35 min
Vehicle weight	–	8.8 kg	–	100 kg	–	6 kg	–	10 kg	–
Number of nozzles	4	4	2	–	4	4	2	4	5
Type of nozzle	–	–	LUI20–02	–	Flat fan 01	Flat fan	Centrifugal atomizer	Electrostatic	2 flat fan and 1 cone
Tank capacity	10 L	10 L	12 L	16 L	15 L	5 L	10 L	10 L	15 L
Rotor diameter	–	–	2.41 m	3.1 m	2.24 m	–	2.22 m	–	2.46 m
Number of rotors	8	8	1	1	1	6	6	6	1
Vehicle length	1471 mm	–	2750 mm	–	420 mm	–	1500 mm	–	–
Vehicle height	482 mm	–	1080 mm	–	450 mm	–	600 mm	–	–
Vehicle width	1471 mm	–	720 mm	–	1300 mm	–	400 mm	–	–
Cost per vehicle	$15,000	$32,000	$120,000	$35,000	–	$13,000	–	$27,000	–

12.4.1 Current Uses of Drones in Agriculture

Currently, drones are utilized in agriculture for a number of functions, which include the following:

12.4.1.1 Crop Monitoring

Drones using cameras and sensors can rapidly and precisely survey fields and give farmers detailed information about the health of their crops, the state of the soil, and any potential issues. This enables farmers to decide on planting, fertilizing, and harvesting based on data, which can boost crop yields and lower expenses.

12.4.1.2 Crop Spraying

As fewer chemicals are needed and the risk for environmental impact is reduced, drones with precision spraying technology can target particular portions of a field.

12.4.1.3 Precision Agriculture

Drones can be utilized for precision planting, which entails placing seeds at precise depths and places to boost crop yields.

12.4.1.4 Mapping

High-resolution maps of fields and farmland may be made with drones, giving farmers comprehensive details about the terrain and soil characteristics of their land.

12.4.1.5 Livestock Monitoring

Drones can be used to track the location, movement, and behavior of livestock to give farmers information about the health and welfare of their animals.

12.4.1.6 Search and Rescue

Drones can be used to find lost pets or find persons who are missing in far-off places. In precision agriculture, which uses technology to maximize crop yield and reduce waste, drones also play a crucial role. The principle behind precision agriculture is that different areas of a field have distinct demands, and these needs can be satisfied by carefully and precisely applying resources like seeds, fertilizer, and water. Drones using cameras and sensors can rapidly and precisely survey fields and give farmers detailed information about the health of their crops, the state of the soil, and any potential issues. Using this data, precise maps of the fields may be made, displaying differences in the soil properties, crop health, and other elements. One of the main uses of drones in precision agriculture is precision planting. Crop yields can be increased by using drones with precision planting systems to place seeds at precise depths and places. In regions where planting by hand is challenging or when the soil is rough or uneven, this can be especially helpful. Overall, drones are turning into a crucial tool in precision agriculture, enabling farmers to choose when to sow, fertilize, and harvest crops based on data, boosting crop yields and cutting expenses. Generally speaking, drones are a crucial tool for farmers, enabling them to increase the effectiveness and profitability of their operations (Debangshi 2021).

12.4.2 Merits of Drones in the Agriculture Sector

In the agricultural industry, drones can be used for a variety of functions, such as crop mapping, soil analysis, irrigation, and pest management. The following are a few of the major advantages of employing drones in agriculture:

12.4.2.1 Increase Effectiveness

Drones can swiftly and efficiently cover huge amounts of land, enabling farmers to collect information and monitor crops more successfully. Early problem identification through this can result in quicker and more efficient responses.

12.4.2.2 Increase Crop Yields

Farmers can spot problem regions by using drones to collect information on the health of their crops. Farmers can raise their agricultural yields and revenues by solving these problems.

12.4.2.3 Lower Expenses

By identifying farm areas that need attention, lowering the demand for physical work, and lowering the use of pesticides and other chemicals, drones can assist to lower prices.

12.4.2.4 Greater Accuracies

Drones can collect precise data and high-resolution photographs, giving farmers a clearer picture of their crops. This can aid in locating problem areas and guarantee that solutions are focused and successful.

12.5 Basics of the UAV Working Methodology

Unmanned aerial vehicles (UAVs) are described as powered machines or aerial vehicles operated autonomously or under remote control, without the need of a human operator. They carry a lethal or nonlethal payload and can be expendable or recoverable. A UAV, often referred to as a drone, unmanned aerial vehicle, or remotely piloted aircraft (RPA), has its operation controlled remotely by a pilot on the ground or in another vehicle or autonomously by onboard computers. The modern-day UAVs are GPS-based systems (Chavan 2019; Hafeez et al. 2022). Frame, motors, propellers, battery, flight controller, cameras, and sensors constitute the basic parts of drone. The frame is a major component of the drone which holds the momentum of the propeller and motors and cameras. A good frame is desired to be sturdy and offer less aerodynamic resistance. Drones are equipped with tiny sensors such as accelerometers, magnetometers, gyros, and pressure sensors as depicted in Fig. 12.2.

The unmanned aerial vehicles work on the principle of aerodynamics or fluid dynamics. There are four major forces acting on the drone, which are responsible for its functioning. Weight acts always in the direction of gravity, lift, thrust, and drag forces. The

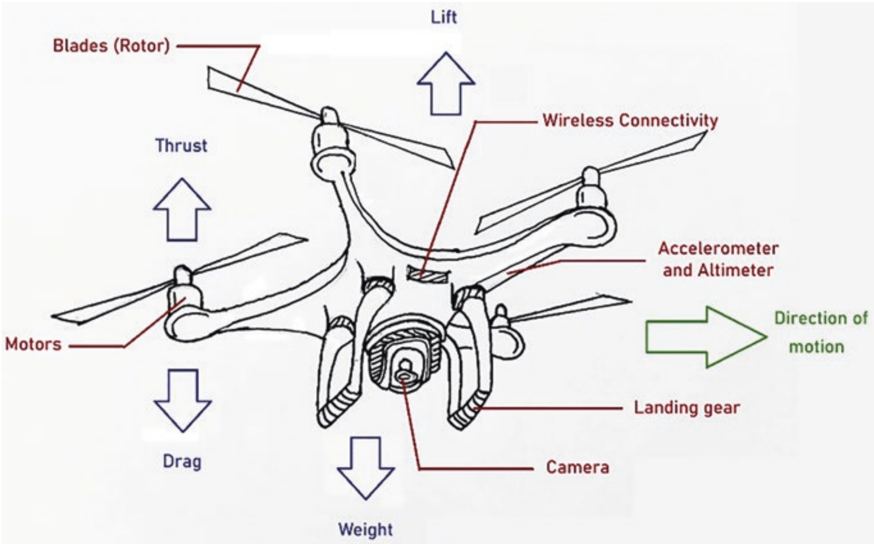

Fig. 12.2 Model diagram of a drone with direction of forces acting on it

drone is lifted against gravity by an upward force called lift resultant of pressure difference across the propeller, i.e., high fluid pressure at the bottom and low fluid pressure at the top of the propeller, which varies with the weight and angle of inclination of the propeller. Force which is responsible for the motion of the drone in a desired direction is called thrust that acts vertically during hovering (Iwasaki et al. 2019).

Opposite to the thrust is the drag force, which acts on the drone in a direction opposite to that of motion resulting from the air resistance. The pressure difference across the propeller and the viscosity of air around the propeller also result in the drag force. Above fixed-wing or helicopter UAVs, in recent times, multicopter UAVs are gaining importance and being widely put to use, of which quadcopter is the most common. The quadcopter drone consists of four propellors mounted on the four corners of the frame. The speed and direction of each of these propellers are individually controlled to maintain the balance and movement of the drone. For the balance, direction of rotation of both the pairs of rotors is different; if one rotates in the clockwise direction, the other rotates in the anticlockwise direction. While the movement of the drone can be achieved by adjusting the speed of the rotor, such as when all rotors run at high speed, the drone hovers, i.e., moves up. The movement of a typical quadcopter can be of four types: throttle, pitch, roll, and yaw. Throttle refers to hovering or upward and downward movement of the drone. Pitch refers to the forward or backward movement of the drone about a lateral axis. This direction depends on the speed of the front and rear propellors. When the rear propellers run at high speed, the drone moves in the forward direction, whereas when the front propellers run at high speed, the drone moves in the backward direction. Roll refers

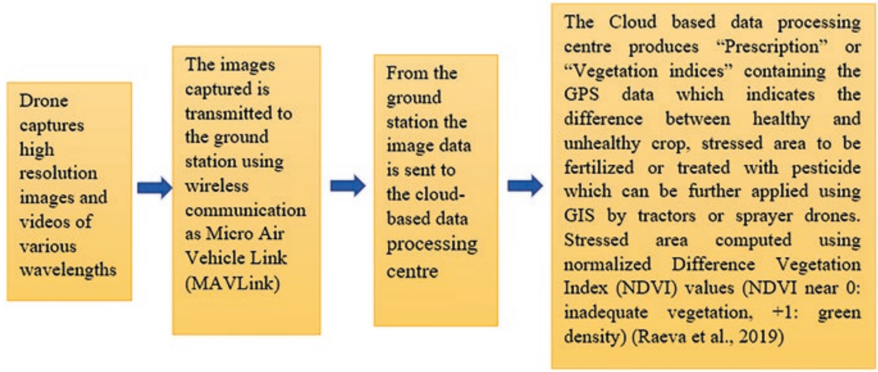

Fig. 12.3 Typical workflow of an agricultural drone

to the movement of the drone in left or right direction about a longitudinal axis. When the right propellors run at high speed, the drone moves in the left direction and vice versa. Yaw refers to the clockwise or anticlockwise rotation of the head of the drone about a vertical axis. Nowadays, focus has shifted to the integration of artificial intelligence (AI) with the drone technology (Raeva et al. 2019; Hafeez et al. 2022). A typical workflow of an agricultural drone is explained in Fig. 12.3.

12.6 Current Drone Technology Available in the Market

Pesticides, as well as fertilizers, are essential elements in pest management and crop growth. Individuals who apply agricultural chemicals and fertilizers by hand get tumors, hypersensitivity, allergies, and other ailments. As a result, drones may be utilized to automate tasks such as fertilizer application, pesticide spraying, and field tracking. In addition, drones may be used for rescue operations, law enforcement, code evaluations, and disaster recovery, including flames. Drones have relatively quick maneuverability, increased cargo capacity, significant pulling power, and stability. This acts as a global nozzle that is capable of spraying both solid and liquid items. A worldwide sprayer distributes insecticides along with fertilizers, while a tension pump sprays only pesticides rather than fertilizers. The global positioning system (GPS) could be utilized to autonomously steer a quadcopter over large expanses. Water-level monitoring sensors operate quadcopters. Based on the above trial, they found that the appropriate application of UAVs in agriculture could reduce water, chemical abuse, and manpower usage by 20–90% (Talaviya et al. 2020; Lost Filho et al. 2020; Borikar et al. 2022). Drone technology currently available in the market is given in Table 12.3.

Table 12.3 Recently developed drone technologies in the globe

Drone name	Company	Origin	Uses
M-18 Dromader	PZL-Mielec	Poland	Aerial application and fire flying
Snow S-2	Ayres Corporation and Thrust Aircraft	American	Aerial application aircraft
Yamaha R-MAX	Yamaha Motor Company	Japan	Aerial surveys, disaster response
AG-272	Hylio	American	High-precision spraying system
Agras T20	SZ DJI Technology Co. Ltd.	Chinese	Crop spraying and dusting
Agras MG-1S	DJI	Chinese	Crop dusting and aerial spraying

12.7 Use of Drone for the Detection of Insect Pests and Spraying of Insecticides

Insect pest infestation detection in the field using traditional methods is expensive and time consuming. It can be difficult to deal with when an extensive area is involved. The arthropod pests are small enough to not be seen with the unaided eye; they reside in the soil or are found in tall trees. The effectiveness of scouting in diverse cropping systems may be hampered by the lack of reliable pest sampling techniques. One of the main drivers of the adoption of drone-based remote sensing technologies in agriculture is the cost of the technology for growers due to the potential time savings from automated crop monitoring, especially for the detection of insect pests (Backoulou et al. 2011; Dara 2019). Advanced image capture and processing power has enabled automatic identification of insect pests, along with the symptoms and degree of damage, as seen in a model proposed by Chouhan et al. (2021a) where identification and classification of midge, *Pauropsylla tuberculata*-induced galls, has been achieved by using an Internet of Things-enabled fuzzy-based function network (IoT FBFN). Drone technology used to detect insect pest infestation in crops and orchards is summarized in Table 12.4.

Drones could help to control pests at these hotspots in addition to being able to identify the pest hotspots. By applying variable insecticides, pest hotspots may be controlled. Since many years ago, pesticides have been sprayed from aircraft, but the products are spread over vast regions and a significant portion is lost to drift (Bird et al. 1996). This is a problem for nearby terrestrial and aquatic ecosystems that have been lost to drift, as well as for human health (Damalas 2015; Chowdhury et al. 2023). Droplet size which is influenced by nozzle type and product composition, meteorological conditions such as wind speed and direction, and application method such as spray height above the canopy are the main variables that affect spray drift (Al Heidary et al. 2014). Therefore, there is a great need for better pesticide application techniques. The use of drones for precise insecticide and miticide

Table 12.4 Use of drone to detect insect pest infestation in major agricultural and horticultural crops and orchards

Plant species	Pest species	Order: family	Monitoring of field	Technology details	Types	Spectral resolution	Sensor details	No. of spectral bands	Reference
Cotton, *Gossypium hirsutum* Linn.	Two-spotted spider mite, *Tetranychus urticae* Koch	Acarina: Tetranychidae	Assessment of damage	Matrice 100, SZ DJI Technology Co.	Four rotors	Multispectral	ADC-Lite, Tetracam Inc.	3	Huang et al. (2018)
Potato, *Solanum tuberosum* Linn.	Colorado potato beetle, *Leptinotarsa decemlineata* (Say)	Coleoptera: Chrysomelidae	Assessment of damage	Spreading Wings S800, SZ DJI Technology Co.	Six rotors	Multispectral	Mini-MCA6, Tetracam Inc.	6	Hunt Jr and Rondon (2017); Hunt Jr et al. (2016)
Sorghum, *Sorghum bicolor* (Linn.) Moench	Sugarcane aphid, *Melanaphis sacchari* (Zehnter)	Hemiptera: Aphididae	Counts of arthropods	eBee, senseFly	Fixed wing	Multispectral	S110 NIRb, Canon	3	Stanton et al. (2017)
Canola, *Brassica* spp.	Green peach aphid, *Myzus persicae* (Sulzer)	Hemiptera: Aphididae	Counts of arthropods	Cinestar-8 MK Heavy Lift, Freefly Systems	Eight rotors	Multispectral	Mini-MCA6, Tetracam Inc.	6	Severtson et al. (2016)
Onion, *Allium cepa* Linn.	Thrips	Thysanoptera: Thripidae	Information not provided	eBee, senseFly	Fixed wing	Multispectral	S110 NIRb, Canon	3	Nebiker et al. (2016)

Grape, *Vitis vinifera* Linn.	Grape phylloxera, *Daktulosphaira vitifoliae*	Hemiptera: Phylloxeridae	Ground traps, and root digging, visual vigor assessments	S800 EVO, SZ DJI Technology Co./Phantom3 Pro, SZ DJI Technology Co.	Six rotors /four rotors	Red blue green + multispectral + hyperspectral	5DsR, Canon Inc. + RedEdge, MicaSense Inc. + NanoHyperspec, Headwall Photonics Inc./Phantom3 Pro associated camera	3 + 5 + 274	Vanegas et al. (2018)
Wheat, *Triticum aestivum* Linn.	Fall armyworm, *Spodoptera frugiperda* Smith	Lepidoptera: Noctuidae	Outbreak reported by grower	Aeryon Scout, Aeryon Labs Inc.	Four rotors	Red blue green + multispectral	Photo3S, Aeryon Labs Inc. + ADCLite, Tetracam Inc.	3 + 3	Zhang et al. (2014)

Table 12.5 A comprehensive detail of drones used for spraying pesticides

Drones	Capacity of tank (L)	Average flight time (min)	Average speed (m/s)	Discharge rate (L/S)	Number of nozzles	Reference
DJI Agrus MG-1S	10	10	12	0.379	4	Rahman et al. (2021)
3WQF120–12	12	30	5	0.8	2	Shilin et al. (2017)
3CD-15	15	20	6	0.54	4	Shilin et al. (2017)
WSZ-0610	10	20	4	0.72	2	Shilin et al. (2017)
HY-B-15 L	15	15	4.5	0.38	5	Shilin et al. (2017)
N-3 UAV	25	N/A	4	0.85	2	Qin et al. (2018)
Knapsack-type electric fog sprayer 3WBD	20	N/A	1	1	1	Qin et al. (2018)

spraying is a possibility. Small drone-based crop dusters, such as the DJI AGRAS MG-1S with a 10-kg payload, have recently entered the market (DJI 2019). In addition to other crops including wheat, oats, and soybeans, unmanned crop dusters are now used on an increasing variety of other crops as well (Yamaha 2016). Major uses of drones for spraying of pesticides (insecticides, herbicides, and fungicides) are shown in Table 12.5.

12.8 Use of Drone for Detection of Disease Infection and Spraying of Fungicides

Similar to insecticides, drones equipped with sensors are used for detection of plant diseases and spraying of fungicides. Earlier satellites were used for mapping and large-scale survey, but advanced sensors and imaging capabilities of UAVs now provide better quality and precision of data, irrespective of weather and positioning (Chouhan et al. 2020). Most UAVs use thermal and multispectral cameras to record the reflectance of vegetation canopy and gather information on different parameters like normalized difference vegetation index (NDVI) and leaf area index (LAI). Every picture taken by the sensor is associated with GPS coordinates. Through advanced image data analytical tools, UAVs are able to detect and distinguish between healthy, nutritionally deficient, or infected crops and also determine the level of infection. Based on these collected data, required amount of fungicides are spread to the particular infected crops living behind the untreated healthy crops (Mogili and Deepak 2018; Dileep et al. 2020).

In large-scale area or commercial farming, drones have already started to replace manual screening, monitoring, and pesticide spray. Advanced processing tools have made possible detection and elimination of a number of diseases over several crops. Higher efficiency, flexibility, high spatial resolution, and quick detection of plant diseases over large area make drone-based remote sensing more effective than other methods for monitoring disease spread, detection, and diagnosis. Digital image processing and soft computing methods are tremendously used for survey of plant diseases (Chouhan et al. 2019a, b). There are different sensors which are used for detecting various types of diseases like RGB digital camera, multispectral camera, hyperspectral sensing, and thermal infrared cameras. Some of the newest models of drones used for disease detection include UAV system S1000 (yellow rust), multirotor DJI Mavic Pro (olive quick decline syndrome), DJI Matrice 600 pro (wheat rust, target spot, and bacterial spot disease), FeimaD200 quadrotor (pine wood nematode disease), etc. (Abbas et al. 2023; Abdulridha et al. 2023). Ziya et al. (2018) determined levels of *Cercospora* leaf spot disease in sugar beet in Tokat, Turkey, by using Image Processing Toolbox module. The disease severity index made using this processing technique was found to be more precise and accurate than manual observation. Heim et al. (2019) used multispectral aerial imaging and unmanned aerial systems (UASs) to detect myrtle rust (*Austropuccinia psidii*) on lemon myrtle plantation. Treated and untreated trees could be classified with 95% accuracy at canopy level with near-infrared (NIR) and red-edge (RE) spectral band. They found that UAS and multispectral camera could be used to detect disease hotspot and pinpoint areas with potential disease incidence. Chouhan et al. (2021b) used a hybrid neural network, incorporated with superpixel clustering to formulate a disease diagnosis system for *Jatropha curcas* and *Pongamia pinnata*. They also proposed a bio-inspired bacterial foraging optimization algorithm and another autonomous disease detection system using radial basis function neural network for segmenting between fungal, bacterial, and viral diseases in plants (Chouhan et al. 2019a, b).

Generally, detection and spray operation go hand in hand. But it is possible to use drones for spraying purpose only, as a low-cost operation. Drones are particularly useful for spraying in terrains or places where a normal laborer would find difficulty in conducting the spray operation. In case of areca nut, it is much difficult and time consuming for a farmer or laborer to climb the tree and then spray the fungicide. To address this problem, a quadcopter drone with Pixhawk Flight Controller and 500-g payload capacity was developed. The use of this drone now allowed even spray of fungicide on each and every tree within a short time period with less labor (Hajare et al. 2021).

12.9 Use of Drone for Detection of Weeds and Spraying of Herbicides

According to Barba et al. (2020), unwanted plants called weeds can lower crop yields by interacting with crops for resources like light, space, water, nutrients, and carbon dioxide. These resources must be managed to meet the needs of food production in the future. Drones, artificially intelligent systems, and a variety of sensors including RGB (red, green, and blue), multispectral, and hyperspectral are integrated to potentially improve the management of weed issues. The majority of the significant or small obstacles brought about by infestations of weeds can be overcome by integrating remote sensing technologies into different agricultural chores (Bini et al. 2020).

With the use of unmanned ground vehicles (UGVs) and drones, this technique can produce food of higher quality while requiring less manpower. De Castro et al. (2018) stated that photographical imagery using drones helps to classify outcomes in early-season agronomic circumstances when crop and seedlings of weeds have comparable spectral signatures. By identifying weeds early in the growing season, precision farming techniques can effectively manage problems with weeds while reducing operating costs and environmental harm. This analysis seeks to provide an overview of the most cutting-edge sensors now on the market for precision weed management.

12.9.1 Application of Drone Technology in Weed Management

Mapping crops or weeds is one of the most effective ways to manage agricultural crops. Du et al. (2008) found that several technological appliances, including global positioning systems (GPSs), remote sensing, variable rate technology (VRT), and geographical information systems (GIS), are integrated into precision agriculture. Applications in the agriculture industry could greatly benefit from the use of remotely sensed data at many different scales, from the microscopic to the global examination of crops that are important globally.

Data from remote sensing systems can take many different forms, ranging from aerial imaging systems using collected data to deployment of ground-based sensors from tractors or other instruments. Shaw (2005) claimed that remote sensing is a desirable tool for identifying agronomic production systems and weed infestations and determining the size and location of weed patches that are common problems, because it can detect minute changes in vegetation features. According to Smith and Blackshaw (2003), remote sensing in photography creates the digital reflection value for each pixel. For example, in weed mapping process, the scene component includes soil, shadow, and various weed and crop species.

12.9.2 Weed Management Through Drone System

Owing to the fact that the field is divided into management zones to prevent weed plants from spreading beyond a small area of the field, each of them is given the previously mentioned tailored management plan. An adequate weed cover map is necessary for targeted spraying of herbicide in order to accomplish this. Today, early-period weed mapping requires a certain level of geographical and temporal resolution, which drones or unnamed aerial vehicles (UAVs) can now deliver (Madsen et al. 2020). According to Wu et al. (2021), in addition to taking photographs with extremely high spatial resolution (pixels as small as a few millimeters or centimeters), UAVs can additionally track smaller individual plants or patches, which was previously not possible. Several researchers demonstrated a specific herbicide for definite location using UAV imagery, and they came to the conclusion that the information on weeds of a specific location could be identified using the imagery. Small unmanned aerial systems (UAS) show great promise for site-specific weed management and are comparatively easy to incorporate into weed research (Xiang and Tian 2011).

It is essential to preserve or expand the variety of herbicides used, as well as to use diverse planting techniques like tillage and crop rotation, which have been shown to lessen weed resistance (Kniss 2018). Unmanned aerial vehicles can be used to identify weed patches and aid in integrated weed management (IWM), which reduces the amount of chemicals released into the environment, and the herbicide-resistant weeds are under selection pressure (López-Granados et al. 2016). AI can therefore assist farmers in managing weeds by detecting them early and preventing future weed resistance. Early weed detection and differentiation in the crop are possible thanks to artificial intelligence (AI), which can also stop weed invasion nearby.

12.10 Challenges and Future Prospects of Drone Technology

12.10.1 Challenges of Drone Use in Agriculture

It has been some years since drones have come into the market. Yet its use is only limited to developed countries or big commercial firms and institutions. This is due to the many shortcomings for which drone use is still restricted in agriculture:

- A conventional farmer is not able to thoroughly study drone photos. For the purpose of operating an agriculture drone, some fundamental skills and knowledge are required.
- Drones could crash with human-operated aviation if they fly in their flight path because they share identical airspace as commercial aircraft. If they lose control for whatever reason, they could crash whereas they linger above the ground.
- They are challenging to fly in extreme weather. For image capture alone, a good amount of sunlight is necessary. The devices may not be able to fly or record the necessary space if there is intense rain, fog, or wind. A drone’s electronic parts are susceptible to damage from rain.
- Given the frequently changing laws and regulations governing drone operation, the expanding list of airspace limits may result in penalties, both monetary and legal.
- A drone survey’s battery life is its major drawback. Short battery life shortens the drone’s flight time, which is disadvantageous for covering large areas or long operations.
- High initial expenses and regulation changes are two of the most difficult obstacles to get over in order to make drone technology well liked and farmer friendly.

12.10.2 Future Scopes Using the Advantages of Drones in Agriculture

In recent years, drone technology in agriculture has advanced significantly. Drones are now more accessible and simpler to use thanks to advancements in technology, which has increased their popularity among farmers. Any crop can be monitored using drones, anywhere. By using drone technology, land management can become more sustainable, crop yields can be increased, and time can be saved.

12.10.2.1 Field Analysis of the Soils

Remote sensing cameras on drones gather information from the ground using electromagnetic spectrum to assess the soil and field. Drones gather raw data, which is then processed using algorithms to produce information that is relevant. As a result, they are appropriate for various farming activities, such as keeping an eye on the following factors:

- Health of crop includes pest-caused damage, nutrient deficits, and changes in color brought on by pest infestation.
- Vegetation catalogues: yield, phenology, leaf area, and efficacy of treatments.
- Growth of plants is measured by their height, LAI, and density.
- Plant assessment: stand quantity, plant size, statistics for the field, a look for compromised fields, and planter skips.
- Water requirements: depending on the environment, the amount of water required will vary, and any moisture-stressed areas of an agricultural land or orchard need to be watered.
- Soil investigation: nutrient availability for plant nutrient control and nutrient concentration in plants.

12.10.2.2 Airborne Seed Sowing

But we can swiftly reforest using drone seeders. In some areas, such as in mangrove forests or steep terrain, these drones' pneumatic firing mechanism drives seed pods deeper into the soil. It can reduce planting expenses by 85% while achieving an emergence rate of 90%.

12.10.2.3 Agricultural Spraying Procedure

Spraying pesticides to get rid of pests and undesired weed-like plants is becoming increasingly important for crop health. Pathak et al. (2020) suggested that the effective farms employ drones for spraying in order to prevent exposing humans to hazardous fertilizers, pesticides, and other chemicals. For jobs like agricultural spraying and planting, GPS technology enables drones to travel to specific sites and maintain a precise flight path. Drones with GPS capabilities can fly in pre-planned patterns over a field, ensuring that every area is treated equally and minimizing the possibility of overlap.

12.10.2.4 Plant Health Evaluation

Drones with crop-scanning sensors employing both visible light and infrared can be utilized to monitor both the condition of crops over time and the efficacy of corrective treatments. This can be trained to recognize characteristics like the NDVI, stress caused by water, or a deficiency of certain nutrients in plants, claimed by Pravin and Munde (2019).

12.10.2.5 Monitoring and Planning for Irrigation

In addition to helping with irrigation-related difficulties, drones with thermal imagery and remote sensing abilities may categorize areas into various moisture regimes. This makes irrigation planning more exact. The FAO drones in the Philippines, as stated by Pravin and Munde (2019), rely on photogrammetric technology and directional technologies having a 3 cm ground-based resolution.

12.10.2.6 Analysis of Plant Emergence and Crop Counts

Unmanned aircraft systems (UAVs) are a practical, quick, and less expensive method for gathering information on crop emergence, assisting with yield prediction. The amount of timber generated in forests can be determined using drones equipped with sensors to determine the alteration in estimating tree/crop biomass based on differences in height.

12.10.2.7 Reduced Risk of Disaster

There have been several approaches made to create procedures for drone-assisted data gathering. Such information can provide rural communities with high-quality, reliable guidance while also helping the government better plan its support and response operations. Instantaneous ground response is made possible by drones, which is much faster than manual detection, analysis, and response.

12.11 Conclusion

The mechanization and modernization of farming practices have widened the scope of drone utilization. In the past decades, drone technology has seen tremendous development, and now there is a wide variety of drones available in a wide price range, with specifications corresponding to requirement of the operation. The use of drones for spraying pesticides has improved overall efficiency and made agriculture a laborless activity, saving time and removing threats to human health from pesticide exposure. In the upcoming times, the use of drones will assist the budding entrepreneurs and lead conventional farmers towards modern agriculture.

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